Polar Coordinates & Polar Graphs

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Analyze the polar curve $r(\theta) = 2 - 4\cos\theta$ for $0 \le \theta < 2\pi$. Identify the curve type, find the maximum distance from the pole, and determine the exact angles θ where the curve passes through the pole.

Step 1: Classify the limaçon $r = a - b\cos\theta$: Since $\frac{a}{b} = \frac{2}{4} = \frac{1}{2} < 1$, the curve is a limaçon with an inner loop. Step 2: Maximum distance from the pole: Occurs when $\cos\theta = -1$ ($\theta = \pi$): $r_{\max} = 2 - 4(-1) = \mathbf{6}$. Step 3: Angles at the pole ($r = 0$): $2 - 4\cos\theta = 0 \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \boldsymbol{\theta = \frac{\pi}{3},\ \frac{5\pi}{3}}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the exact angle θ between $\mathbf{u} = \langle 3, 1\rangle$ and $\mathbf{v} = \langle 1, 2\rangle$, and compute the vector projection $\text{proj}_{\mathbf{v}}\mathbf{u}$.

Step 1: Compute dot product and magnitudes: $\mathbf{u}\cdot\mathbf{v} = 3(1) + 1(2) = 5, \quad ||\mathbf{u}|| = \sqrt{10}, \quad ||\mathbf{v}|| = \sqrt{5}$. Step 2: Angle between vectors: $\cos\theta = \frac{5}{\sqrt{10}\sqrt{5}} = \frac{5}{5\sqrt{2}} = \frac{\sqrt{2}}{2} \Rightarrow \boldsymbol{\theta = \frac{\pi}{4} = 45^\circ}$. Step 3: Vector projection onto $\mathbf{v}$.

3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A cup of coffee at 92°C is placed in a room kept at a constant 20°C. After 10 minutes, the coffee has cooled to 68°C. Using Newton's Law of Cooling $T(t) = 20 + 72b^{t/10}$, find the exact temperature after 20 minutes and when the coffee reaches 36°C.

Step 1: Find the 10-minute decay base b: $T(10) = 20 + 72b^1 = 68 \Rightarrow 72b = 48 \Rightarrow b = \frac{48}{72} = \frac{2}{3}$. Step 2: Temperature at $t = 20$ minutes: $T(20) = 20 + 72\left(\frac{2}{3}\right)^{20/10} = 20 + 72\left(\frac{4}{9}\right) = 20 + 32 = \mathbf{52}^\circ$C. Step 3: Solve $T(t) = 36^\circ$C: $20 + 72\left(\frac{2}{3}\right)^{t/10} = 36 \Rightarrow \left(\frac{2}{3}\right)^{t/10} = \frac{16}{72} = \frac{2}{9}$. .

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Given that $\cos\alpha = -\frac{5}{13}$ with $\frac{\pi}{2} < \alpha < \pi$, find the exact values of $\sin(2\alpha)$, $\cos(2\alpha)$, and $\tan(2\alpha)$.

Step 1: Find $\sin\alpha$ in Quadrant II ($\sin\alpha > 0$): $\sin\alpha = \sqrt{1 - \left(-\frac{5}{13}\right)^2} = \sqrt{1 - \frac{25}{169}} = \frac{12}{13}$. Step 2: Apply double-angle formulas: $\sin(2\alpha) = 2\sin\alpha\cos\alpha = 2\left(\frac{12}{13}\right)\left(-\frac{5}{13}\right) = \boldsymbol{-\frac{120}{169}}$.
$\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = \frac{25}{169} - \frac{144}{169} = \boldsymbol{-\frac{119}{169}}$. $\tan(2\alpha) = \frac{\sin(2\alpha)}{\cos(2\alpha)} = \boldsymbol{\frac{120}{119}}$.

5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Evaluate the exact value of $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$ and $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$.

Step 1: Evaluate $\sin\left(\arccos\left(-\frac{3}{5}\right)\right)$: Let $\theta = \arccos(-3/5) \in [0, \pi]$. Since $\cos\theta = -3/5 < 0$, θ is in Quadrant II where $\sin\theta > 0$: $\sin\theta = \sqrt{1 - (-3/5)^2} = \boldsymbol{\frac{4}{5}}$. Step 2: Evaluate $\tan\left(\arcsin\left(\frac{8}{17}\right)\right)$: Let $\phi = \arcsin(8/17) \in (0, \pi/2)$. Then $\cos\phi = \sqrt{1 - (8/17)^2} = \frac{15}{17}$, so: $\tan\phi = \frac{8/17}{15/17} = \boldsymbol{\frac{8}{15}}$.

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6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For a quadratic function $f(x)$, selected values are $f(1) = 4$, $f(2) = 9$, $f(3) = 18$, and $f(4) = 31$. Compute the first and second differences, verify that f is quadratic, and find its explicit formula $f(x) = ax^2 + bx + c$.

Step 1: Compute first and second differences:First differences Δy: $9 - 4 = 5$, $18 - 9 = 9$, $31 - 18 = 13$ .Second differences $\Delta^2 y$: $9 - 5 = 4$, $13 - 9 = 4$.Since $\Delta^2 y = 4$ is constant, $f(x)$ is quadratic with $2a = 4 \Rightarrow a = 2$. Step 2: Solve for b and c:$f(1) = 2(1)^2 + b(1) + c = 4 \Rightarrow b + c = 2$. $f(2) = 2(2)^2 + b(2) + c = 9 \Rightarrow 2b + c = 1$.Subtracting gives $b = -1$ and $c = 3$. Thus $\mathbf{f(x) = 2x^2 - x + 3}$.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let $f(x) = \frac{4e^{2x}}{e^{2x} + 3}$. Prove that f is strictly one-to-one and find an explicit formula for its inverse function $f^{-1}(x)$, stating the domain of f^{-1}.

Step 1: Solve $y = \frac{4e^{2x}}{e^{2x} + 3}$ for e^{2x}: $y(e^{2x} + 3) = 4e^{2x} \Rightarrow 3y = e^{2x}(4 - y) \Rightarrow e^{2x} = \frac{3y}{4 - y}$. Step 2: Take the natural logarithm and divide by 2: $2x = \ln\left(\frac{3y}{4 - y}\right) \Rightarrow f^{-1}(x) = \mathbf{\frac{1}{2}}\ln\left(\mathbf{\frac{3x}{4 - x}}\right)$. Step 3: Domain of f^{-1} (range of f): Since $e^{2x} > 0$, we require $\frac{3x}{4 - x} > 0 \Rightarrow \mathbf{0 < x < 4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Identify the conic section $4x^2 - 9y^2 - 16x - 18y - 29 = 0$ by completing the square. Find its center, vertices, foci, and asymptotes.

Step 1: Group and complete the square: $4(x^2 - 4x + 4) - 9(y^2 + 2y + 1) = 29 + 16 - 9 = 36$. $\mathbf{\frac{(x-2)^2}{9} - \frac{(y+1)^2}{4} = 1}$. Step 2: Extract hyperbola properties ($a = 3$, $b = 2$, $c = \sqrt{9 + 4} = \sqrt{13}$):Center: $(2, -1)$Vertices: $(-1, -1)$ and $(5, -1)$Foci: $(2 \pm \sqrt{13}, -1)$Asymptotes: $y + 1 = \pm\frac{2}{3}(x - 2)$

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the logarithmic equation $\log_4(x + 6) + \log_4(x - 6) = 3$, checking for extraneous roots.

Step 1: Domain restriction: We require $x + 6 > 0$ and $x - 6 > 0$, so $x > 6$. Step 2: Condense logarithms using the product rule: $\log_4((x + 6)(x - 6)) = 3 \Rightarrow x^2 - 36 = 4^3 = 64$. Step 3: Solve for x and filter by domain: $x^2 = 100 \Rightarrow x = \pm 10$. Reject $x = -10$ since $-10 ot > 6$. The unique valid solution is $\mathbf{x = 10}$.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\log_2(5) \cdot \log_5(9) \cdot \log_3(16)$ without a calculator using the change-of-base formula.

Step 1: Convert all logarithms to natural logarithms: $\frac{\ln 5}{\ln 2} \cdot \frac{\ln 9}{\ln 5} \cdot \frac{\ln 16}{\ln 3}$. Step 2: Simplify using prime power identities $\ln 9 = 2\ln 3$ and $\ln 16 = 4\ln 2$: $\frac{\ln 5}{\ln 2} \cdot \frac{2\ln 3}{\ln 5} \cdot \frac{4\ln 2}{\ln 3} = 2 \times 4 = \mathbf{8}$.

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11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Let $f(x) = 2x^2 - 3$ and $g(x) = \sqrt{x+5}$. Find the composite function $(f \circ g)(x)$ and state its exact domain.

Step 1: Determine the domain of the inner function $g(x)$: $x + 5 \geq 0 \Rightarrow x \geq -5$. Step 2: Substitute $g(x)$ into $f(x)$: $(f \circ g)(x) = f(\sqrt{x+5}) = 2(\sqrt{x+5})^2 - 3 = 2(x+5) - 3 = \mathbf{2x+7}$, with domain $\mathbf{[-5, \infty)}$.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

Convert the polar coordinate point $P\left(6, \frac{4\pi}{3}\right)$ to exact Cartesian coordinates (x, y).

Step 1: Apply conversion formulas: $x = r\cos\theta = 6\cos\left(\frac{4\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$.
$y = r\sin\theta = 6\sin\left(\frac{4\pi}{3}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. The Cartesian coordinates are $(-3, -3\sqrt{3})$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \leq \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Convert the Cartesian equation $(x^2 + y^2)^2 = 4(x^2 - y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2 + y^2 = r^2$, $x = r\cos\theta$, and $y = r\sin\theta$:
$(r^2)^2 = 4(r^2\cos^2\theta - r^2\sin^2\theta) \Rightarrow r^4 = 4r^2(\cos^2\theta - \sin^2\theta)$. Apply the double-angle identity $\cos^2\theta - \sin^2\theta = \cos(2\theta)$: $r^4 = 4r^2\cos(2\theta)$. Divide through by r^2 (for $req0$): $r^2 = 4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r = 2$ at.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 1.03 Rates of Change & Concavity**

Convert the polar equation $r = 8\sin\theta$ into rectangular coordinates, and find its center and radius.

Multiply both sides by r: $r^2 = 8r\sin\theta$ Substitute $r^2 = x^2 + y^2$ and $y = r\sin\theta$:
$x^2 + y^2 = 8y \Rightarrow x^2 + (y^2 - 8y + 16) = 16 \Rightarrow \mathbf{x^2 + (y-4)^2 = 16}$ This is a circle with center $\mathbf{(0, 4)}$ and radius $\mathbf{r = 4}$.

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16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 2.01 Exponential Functions & Growth Models**

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 2.02 Logarithmic Functions & Properties**

Convert the polar equation $r = 4\cos\theta$ to Cartesian rectangular form, classify the curve, and find its center and radius.

Step 1: Multiply both sides by r: $r^2 = 4r\cos\theta$. Step 2: Substitute $r^2 = x^2 + y^2$ and $r\cos\theta = x$: $x^2 + y^2 = 4x \Rightarrow x^2 - 4x + y^2 = 0$. Step 3: Complete the square on x: $(x^2 - 4x + 4) + y^2 = 4 \Rightarrow (x - 2)^2 + y^2 = 4$. This represents a circle with center $(2, 0)$ and radius $r = 2$.

18 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)** • **Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation θ, standard form in the rotated system (x', y'), and the lengths of the semi-major and semi-minor axes.

Step 1: Determine the rotation angle θ: Here $A = 5$, $B = -4$, and $C = 8$. $\cot(2\theta) = \frac{A - C}{B} = \frac{5 - 8}{-4} = \frac{-3}{-4} = \frac{3}{4} > 0$. Since $\cot(2\theta) = 3/4$, by right triangle trigonometry $\cos(2\theta) = \frac{3}{5}$. Using half-angle formulas for $\theta \in (0, \pi/4)$: $\cos\theta = \sqrt{\frac{1 + \cos(2\theta)}{2}} = \sqrt{\frac{1 + 3/5}{2}} = \sqrt{\frac{4}{5}} = \frac{2}{\sqrt{5}},$.

19 **[Challenging]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)** • **Cross-Lesson: 4.01 Inverse & Composite Functions**

Determine the slant asymptote of $g(x) = \frac{2x^3 - 3x^2 + 5}{x^2 + 1}$ and find the coordinates of any point where the curve intersects this asymptote.

Since $\deg(\text{num}) = 3$ and $\deg(\text{denom}) = 2$ with $3 - 2 = 1$, a slant asymptote exists. Perform polynomial long division: $(2x^3 - 3x^2 + 5) \div (x^2 + 1) = (2x - 3) + \frac{-2x + 8}{x^2 + 1}$. The linear quotient gives the slant asymptote: $y = 2x - 3$. To find where the graph intersects this asymptote, set the remainder equal to zero: $\frac{-2x + 8}{x^2 + 1} = 0 \Rightarrow -2x + 8 = 0 \Rightarrow x = 4$. Substitute $x = 4$ into the asymptote equation: $y = 2(4) - 3 = 5$. The curve intersects.

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20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Find the inverse function $f^{-1}(x)$ and state its domain and range for $f(x) = \frac{3x-1}{2x+5}$.

Write $y = \frac{3x-1}{2x+5}$. The domain of f is $x \neq -2.5$. Its horizontal asymptote is $y = 3/2$, so the range of f is $y \neq 1.5$. Swap x and y and solve for y: $x = \frac{3y-1}{2y+5} \Rightarrow x(2y+5) = 3y-1$.
$2xy + 5x = 3y - 1 \Rightarrow 2xy - 3y = -5x - 1$. $y(2x-3) = -(5x+1) \Rightarrow y = \frac{-5x-1}{2x-3} = \frac{5x+1}{3-2x}$. Inverse function: $f^{-1}(x) = \frac{5x+1}{3-2x}$. Domain of f^{-1}: $x \neq \frac{3}{2}$ (the range of f). Range of f^{-1}.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar point $(r, \theta) = \left(6, \frac{5\pi}{6}\right)$ to rectangular coordinates (x, y).

a. $(-3\sqrt{3}, 3)$ ✓

b. $(3\sqrt{3}, -3)$

c. $(-3, 3\sqrt{3})$

d. $(3, -3\sqrt{3})$

$x = r\cos\theta = 6\cos\left(\frac{5\pi}{6}\right) = 6\left(-\frac{\sqrt{3}}{2}\right) = -3\sqrt{3}$. $y = r\sin\theta = 6\sin\left(\frac{5\pi}{6}\right) = 6\left(\frac{1}{2}\right) = 3$. The rectangular point is $(-3\sqrt{3}, 3)$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular point $(-4, 4)$ to polar coordinates with $r > 0$ and $0 \leq \theta < 2\pi$.

a. $\left(4\sqrt{2}, \frac{3\pi}{4}\right)$ ✓

b. $\left(4\sqrt{2}, \frac{7\pi}{4}\right)$

c. $\left(8, \frac{3\pi}{4}\right)$

d. $\left(4\sqrt{2}, \frac{5\pi}{4}\right)$

$r = \sqrt{(-4)^2 + 4^2} = \sqrt{32} = 4\sqrt{2}$. Since $(-4, 4)$ is in Quadrant II, $\theta = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$.

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Which polar point represents the same location as $(r, \theta) = (3, \frac{\pi}{3})$?

a. $(-3, \frac{4\pi}{3})$ ✓

b. $(-3, \frac{\pi}{3})$

c. $(3, -\frac{\pi}{3})$

d. $(-3, \frac{2\pi}{3})$

In polar coordinates, $(r, \theta) = (-r, \theta + \pi)$. For $(3, \frac{\pi}{3})$, $(-3, \frac{\pi}{3} + \pi) = (-3, \frac{4\pi}{3})$ represents the exact same physical point.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the polar equation $r = 6\cos\theta$ into rectangular coordinates.

a. $(x-3)^2 + y^2 = 9$ ✓

b. $x^2 + (y-3)^2 = 9$

c. $x^2 + y^2 = 6x$

d. $x^2 + y^2 = 36$

Multiply both sides by r: $r^2 = 6r\cos\theta$. Substitute $r^2 = x^2 + y^2$ and $x = r\cos\theta$:

$x^2 + y^2 = 6x \Rightarrow x^2 - 6x + 9 + y^2 = 9 \Rightarrow (x-3)^2 + y^2 = 9$ (circle centered at $(3, 0)$ of radius 3).

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert the rectangular equation $x^2 + y^2 = 16$ to polar form.

a. $r = 4$ ✓

b. $r = 16$

c. $r = 4\cos\theta$

d. $r = 16\sin\theta$

Since $r^2 = x^2 + y^2$, we have $r^2 = 16 \Rightarrow r = 4$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $r = 4\csc\theta$ to rectangular coordinates.

a. $y = 4$ ✓

b. $x = 4$

c. $y = 4x$

d. $x^2 + y^2 = 16$

Multiply both sides by $\sin\theta$: $r\sin\theta = 4$. Since $y = r\sin\theta$, this is the horizontal line $y = 4$.

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27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How many petals does the rose curve $r = 5\cos(4\theta)$ have?

a. 8 ✓

b. 4

c. 16

d. 2

For a rose curve $r = a\cos(n\theta)$, if n is even, there are $2n$ petals. Here $n = 4$ is even, so the rose has $2(4) = 8$ petals.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the maximum distance from the pole (origin) to the curve $r = 4\sin(3\theta)$?

a. 4 ✓

b. 12

c. 3

d. 8

The maximum value of $|\sin(3\theta)|$ is 1. Therefore, the maximum radius is $|r| = 4(1) = 4$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Classify the polar curve $r = 2 + 2\cos\theta$.

a. **Cardioid (heart-shaped limaçon)** ✓

b. Limaçon with an inner loop

c. Dimpled limaçon

d. Convex limaçon

For $r = a \pm b\cos\theta$, when $\frac{a}{b} = 1$, the curve is a cardioid.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Which polar equation produces a limaçon with an inner loop?

a. $r = 2 + 4\cos\theta$ ✓

b. $r = 4 + 2\cos\theta$

c. $r = 3 + 3\sin\theta$

d. $r = 5 + 4\cos\theta$

A limaçon $r = a + b\cos\theta$ has an inner loop when $\frac{a}{b} < 1$. For $r = 2 + 4\cos\theta$, $\frac{2}{4} = 0.5 < 1$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

At what angles $\theta \in [0, 2\pi)$ does the curve $r = 2 + 4\cos\theta$ pass through the pole ($r = 0$)?

a. $\theta = \frac{2\pi}{3}$ **and** $\theta = \frac{4\pi}{3}$ ✓

b. $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$

c. $\theta = \frac{\pi}{2}$ and $\theta = \frac{3\pi}{2}$

d. $\theta = \pi$

Set $r = 0 \Rightarrow 2 + 4\cos\theta = 0 \Rightarrow \cos\theta = -\frac{1}{2}$. In $[0, 2\pi)$, this occurs at $\theta = \frac{2\pi}{3}$ and $\theta = \frac{4\pi}{3}$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$ ✓

d. $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer k. From $r^3 = 64$, we get $r = 4$ (since r is a non-negative real number). Fro.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

We are given two complex numbers in polar form: $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. We are provided with $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. For z_2, we are given $r_2 = 2$. The product of two complex numbers in polar form is given by $z_1 \cdot z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$. We are given that $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$, where $\alpha = \frac{5\pi}{6}$. Equating the magnitudes, we.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. **4** ✓

c. $2\sqrt{2}$

d. 1

We are given $z^3 = 8i$. In polar form, $8i = 8(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$. Let $z = r(\cos\theta + i\sin\theta)$. Then $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. Equating the magnitudes: $r^3 = 8$, so $r = 2$. Equating the arguments: $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k. Since z is in the first quadrant, $0 < \theta < \frac{\pi}{2}$. For $k = 0$, $3\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{6}$. This lies in the first quadrant. For $k = 1$.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. **2** ✓

b. 4

c. 8

d. 16

We are given the complex number $z = r(\cos\theta + i\sin\theta)$, where $r > 0$. We are also given that $z^3 = -8$ and the principal argument of z is $\theta = \frac{\pi}{3}$. Using De Moivre's Theorem, we have $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. We are given $z^3 = -8$, which can be written in polar form as $8(\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi))$ for any integer k. Equating the magnitudes, we get $r^3 = 8$. Since $r > 0$, we have $r = \sqrt[3]{8} = 2$. Equating the arguments, we get.

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37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$ ✓

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

The rectangular form of a complex number given in polar form $z = r(\cos(\theta) + i\sin(\theta))$ is $a + bi$, where $a = r\cos(\theta)$ and $b = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{3}$: $a = 5\cos(\frac{\pi}{3}) = 5 \times \frac{1}{2} = \frac{5}{2}$
$b = 5\sin(\frac{\pi}{3}) = 5 \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$ Therefore, the rectangular form is $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. **5** ✓

b. 7

c. 25

d. 16

The complex number can be written as $z = 3 + 4i$. The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. In this case, $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.01 Inverse & Composite Functions

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$ ✓

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

Applying Polar Coordinates principles, Squaring $r = 1$: $r^2 = x^2 + y^2 = 1$. Therefore, the verified correct option is $x^2 + y^2 = 1$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

The polar equation $r = 2\cos\theta$ represents a circle. What is its radius?

a. 2

b. 3

c. 5

d. **1** ✓

$r = 2a\cos\theta$ is the circle $(x - a)^2 + y^2 = a^2$ with $2a = 2 \Rightarrow a = 1$; radius $= 1$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.03 Parametric Equations & Modeling

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. **There exists** $c \in (1, 4)$ **with** $f(c) = 0$ ✓

c. $f(2) = 1$

d. f is constant

IVT: continuous function on $[a, b]$ attains every value between $f(a)$ and $f(b)$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Find θ (in $[0^\circ, 360^\circ)$) for the Cartesian point $(3, 0)$.

a. 180

b. **0** ✓

c. 181

d. 90

The point $(3, 0)$ lies at angle $\theta = 0^\circ$ from the positive x-axis.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Rational Functions & Asymptotes

Convert the polar point $(4, 30^\circ)$ to Cartesian coordinates.

a. $(2\sqrt{3}, 3)$

b. $(-2\sqrt{3}, -2)$

c. $(1 + 2\sqrt{3}, 2)$

d. $(2\sqrt{3}, 2)$ ✓

$x = r\cos\theta = 4\cos 30^\circ = 2\sqrt{3}$, $y = r\sin\theta = 4\sin 30^\circ = 2$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Rates of Change & Concavity

How many petals does the rose $r = \cos(2\theta)$ have?

a. **4** ✓

b. 6

c. 2

d. 5

For $r = \cos(n\theta)$, $n = 2$ is even, so the number of petals is $2n = 4$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Find the area of the polar region bounded by $r = 2$ for $0 \le \theta \le \frac{\pi}{6}$.

a. $\frac{2\pi}{3}$

b. $\frac{\pi}{3}$ ✓

c. $\frac{\pi}{6}$

d. $\frac{4\pi}{3}$

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(2)^2(\frac{\pi}{6}) = \frac{\pi}{3}$.

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